

Developing a City-Scale Energy System Model: Focusing on Building Systems for Sustainable Urban Energy Management

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Abstract—This study develops a city-scale energy system model, aiming to address the growing concerns over climate change and greenhouse gas emissions. Focused on building systems, which account for approximately 40% of global energy consumption and carbon emissions, the model employs Python and EnergyPlus for simulation and prediction. By narrowing the scope to the city level, the model leverages regional weather data and diverse occupant behaviors to provide actionable insights for sustainable urban energy management. It facilitates policy development by setting CO₂ emission targets and suggesting measures to achieve them. While the model offers significant advantages, including improved accuracy at the city scale and support for smart city development, certain limitations persist. These include the exclusion of industrial and office buildings, the focus on operational emissions without considering embodied emissions, and the lack of future weather data. Additionally, real-world complexities such as policy shifts and global energy innovations are challenging to quantify and predict with simple formulas. To address these constraints, emerging technologies like machine learning and digital twins are proposed for future iterations, enabling more precise simulations and predictions through real-time updates and historical data analysis. This approach underscores the model's potential for optimization and its role in advancing sustainable energy solutions.

Keywords—City-scale energy model, Building systems, Sustainable energy solutions, Carbon emissions, Climate change, Simulation and prediction

I. INTRODUCTION AND BACKGROUND

This study aims to develop a city-scale energy system model as a crucial tool for advancing sustainable urban energy management. Cities account for a significant share of global energy consumption and carbon emissions, with buildings alone responsible for approximately 40% of the total energy use [1]. The urgency to address climate change has prompted researchers and policymakers to explore more efficient and sustainable energy solutions, particularly within urban environments [2,3].

The proposed model seeks to enhance the understanding of energy consumption patterns in cities, providing a structured framework to analyze and optimize energy use at a granular level.

A comprehensive city-scale model requires extensive data inputs, including detailed information on energy demand, renewable energy integration, and the impact of various policy interventions [4]. Unlike national-scale models, which aggregate broad energy trends, a city-scale approach allows for the inclusion of regional weather patterns, diverse building typologies, and variations in occupant behavior. By employing Python and EnergyPlus for simulation and prediction, the model aims to generate high-resolution insights that support decision-making processes related to sustainable energy transitions.

Given the increasing emphasis on smart city initiatives, urban energy system modeling plays a vital role in shaping future energy strategies. The developed model will focus on building systems, leveraging real-world data to assess energy performance, carbon emissions, and the feasibility of renewable energy integration [5,6]. By narrowing the study scope to urban environments, this research contributes to the growing body of work on sustainable city planning, offering practical solutions for energy efficiency improvements and emissions reduction.

This project builds upon prior research in multi-energy system modeling and smart city frameworks [7,8]. It acknowledges existing limitations in energy modeling, such as the exclusion of embodied emissions and challenges in predicting long-term energy trends due to policy shifts and technological advancements. Nevertheless, emerging approaches like machine learning and digital twins present opportunities to enhance model accuracy by incorporating real-time data and predictive analytics [9]. This study aims to bridge the gap between theoretical energy system modeling and practical urban energy management, ultimately supporting the transition toward low-carbon, energy-efficient cities.

II. METHODOLOGY

The development of a city-scale energy system model requires a structured approach to accurately simulate energy demand, generation, and emissions. This study follows a three-phase methodology: (1) Individual Model Simulation, (2) Regional Energy Demand and Generation Estimation, and (3) Future Prediction. Each phase incorporates computational techniques that enhance precision and applicability in urban energy management.

A. Individual Model Simulation

To obtain a relatively precise estimation of building energy consumption, a discrete building energy model is initially developed using EnergyPlus. The model simulates the thermal and electrical performance of various building typologies, incorporating heating, ventilation, and air conditioning (HVAC) systems, lighting, and occupant schedules. The flexibility of the model allows for the modification of key parameters, such as building typology, HVAC system classification, and occupant behavior, ensuring comprehensive coverage of different urban contexts.

Additionally, renewable energy sources such as solar photovoltaic (PV) panels and wind turbines are integrated into the model to assess their contribution to building-level energy generation. By implementing diverse regional weather datasets, the model gathers data on power generation under different climatic conditions. The model simulation's key outputs include electricity and natural gas consumption per day throughout the year, as shown in Figure 1.

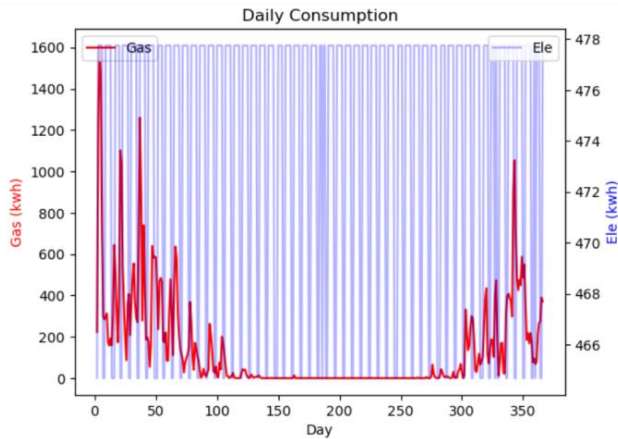


Figure 1. Electricity and natural gas consumption per day, 365 days a year

The energy demand simulation results also form the basis for constructing a Monthly Energy Consumption Chart, which visualizes variations in energy use over different seasons. By analyzing the chart (as shown in Figure 2), trends in peak and off-peak energy consumption can be identified, providing insights into potential energy efficiency improvements.

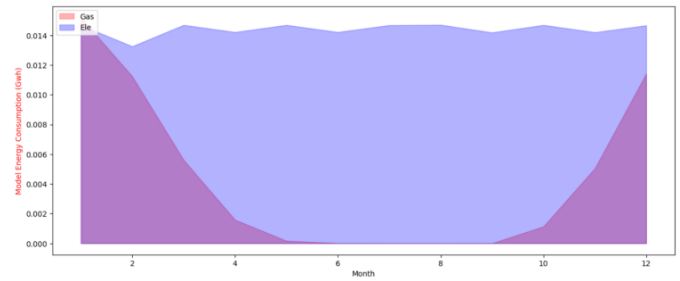


Figure 2. Model Monthly Energy Consumption Chart

B. Regional Energy Demand and Generation Estimation

After acquiring energy consumption data from individual building simulations, the next step involves aggregating the results to estimate city-wide energy demand. This requires integrating data on the number of buildings in each category, the total built area, and population density. As depicted in the Modeled monthly solar and wind energy consumption maps (as shown in Figure 3), different cities exhibit varying levels of renewable energy availability, directly impacting the feasibility of integrating solar and wind power into the urban energy mix.

District Heating is employed to meet thermal energy demand, given that gas heating has a high CO₂ emission factor [10]. The model evaluates the potential of alternative heating methods, including biomass, geothermal energy, and Combined Heat and Power (CHP) systems. CHP plays a crucial role in reducing carbon emissions, as it simultaneously generates electricity and heat with greater efficiency compared to conventional fossil fuel combustion.

Electricity consumption is addressed through a combination of renewable energy generation and conventional power sources. Table 1 provides an overview of different energy contributions, including hydropower, nuclear energy, wind energy (WT), and photovoltaic solar energy (PV). Any shortfall in renewable energy generation is compensated by fuel-based electricity, which significantly impacts CO₂ emissions.

Following the energy demand and supply calculations, total CO₂ emissions are estimated based on emission factors associated with different energy sources. Table 2 summarizes these calculations, highlighting the dominant role of fuel-based electricity in carbon emissions.

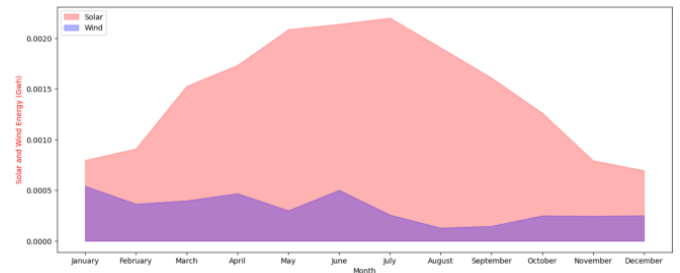


Figure 3. Modeled monthly solar and wind energy consumption maps

Based on the above figure and fuel-based power calculations, the following conclusions are drawn:

TABLE I. REGIONAL ENERGY PRODUCTION, DEMAND, AND CO₂ EMISSIONS STATISTICS (2023)

	Hydro (Gwh)	Nuclear (Gwh)	WT_region (Gwh)	PV_region (Gwh)	Total Demand (Gwh)	Fuel Based (Gwh)
2023	0.374591	1.123774	0.385043	0.993467	3.745913	0.869038

Annotation:WT means The energy produced by wind energy.PV means The energy generated by photovoltaics.Hydro means The energy generated by hydropower.Nuclear means The energy generated by nuclear energy

Calculation of CO₂ emissions :

The following conclusions were drawn based on the common coefficients of natural gas emission factor: 0.18 kg/kwh, fuel-based electricity emission factor: 0.35 kg/kwh, and cogeneration emission factor: 0.224 kg/kWh.

TABLE II. CO₂ EMISSIONS FROM COGENERATION (CHP) AND FUEL-BASED ELECTRICITY (2023)

	CHP_CO2(kg)	Ele_Fuel_CO2(kg)	Total CO2 Emission(kg)
2023	17826.64897	304163.323614	321989.972584

C. Future Prediction

Once the current energy demand and CO₂ emissions are established, the model employs an exponential forecasting approach to predict future energy trends. The prediction considers several key factors:

- Renewable energy growth rates: The anticipated expansion of solar, wind, and other renewable sources.
- Building energy efficiency improvements: The impact of retrofitting, smart controls, and high-performance materials.
- Policy-driven changes: Government regulations influencing energy efficiency standards and renewable energy adoption.
- Climate variability: The effect of temperature fluctuations on heating and cooling demand.

Figure 4 illustrates expected energy contributions over time, providing insights into how renewable energy sources can offset fossil fuel reliance. Additionally, Figure 5 demonstrates the impact of different energy transition pathways on emission reduction goals.

The model's future prediction module, while foundational, requires augmentation to address dynamic urban energy landscapes. First, integrating digital twin technology would enable real-time synchronization between physical infrastructure and virtual simulations. By embedding IoT sensors across building systems (e.g., smart meters, occupancy

detectors), the twin could continuously update parameters such as HVAC efficiency degradation, rooftop PV output fluctuations, and occupant behavior shifts [11]. For example, Helsinki's Energy and Climate Atlas utilizes a city-scale digital twin to simulate district heating scenarios under varying climate projections, achieving a 12% reduction in annual grid strain through adaptive load balancing [12].

Second, the model's exponential forecasting approach should be supplemented with probabilistic scenario analysis to account for policy and technological uncertainties. Monte Carlo simulations could evaluate emission pathways under diverse assumptions—such as renewable subsidy phase-outs, breakthroughs in battery storage, or extreme weather frequency—providing policymakers with risk-weighted recommendations. For instance, a 2040 scenario assuming 5% annual growth in solar PV adoption versus 3% growth would yield distinct grid stability profiles, guiding infrastructure investment priorities.

Third, incorporating high-resolution climate projections is critical for long-term accuracy. Collaborations with meteorological agencies to access CMIP6 (Coupled Model Intercomparison Project Phase 6) datasets would allow the model to simulate energy demand under Representative Concentration Pathways (RCPs) 4.5 and 8.5 [13]. Such integration would reveal, for example, how rising summer temperatures in subtropical cities could amplify cooling demand by 25–40% by 2040, necessitating earlier deployment of passive cooling technologies or district-level thermal storage.

Lastly, demand-side flexibility mechanisms, such as time-of-use pricing and vehicle-to-grid (V2G) integration, should be modeled to assess their impact on peak shaving and renewable curtailment [14]. For example, California's Flex Alert program reduced peak electricity demand by 7% during heatwaves by incentivizing load shifting, a strategy replicable in urban models through agent-based simulations of consumer responsiveness.

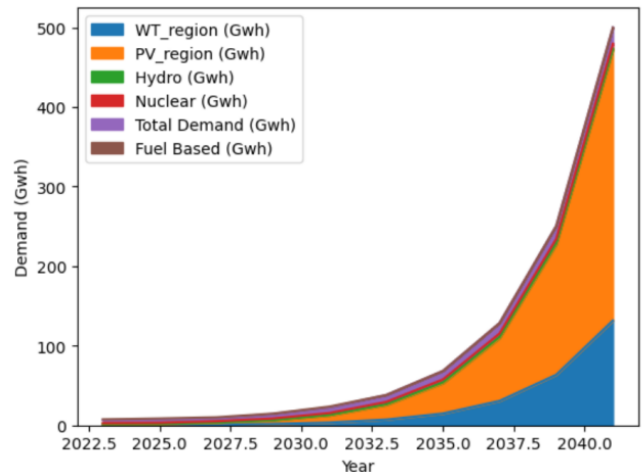


Figure 4. Forecasted energy for wind, solar, nuclear, hydropower, and district energy demand for building systems for each year from 2023 to 2041

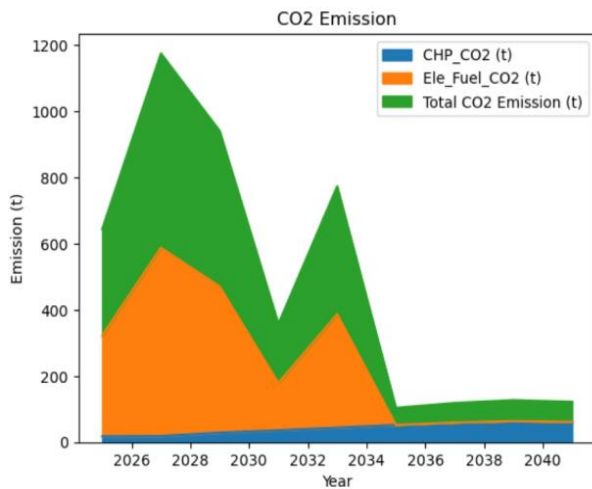


Figure 5. CO2 cogeneration coverage, CO2 emissions released from fuel conversion to electricity and total CO2 emissions from 2023 to 2041

III. ANALYSIS AND CONCLUSION

In summary, this model presents both advantages and limitations. On the positive side, it offers the feasibility of obtaining relatively accurate results by narrowing the focus to a city scale, utilizing distinct regional weather data, and incorporating varying occupant behaviours and equipment to simulate diverse buildings under various scenarios. Furthermore, it can aid in policy development by setting specific targets for controlling CO2 emissions and identifying necessary changes to achieve those goals. Overall, the model provides ample opportunities for further optimisation.

Nonetheless, there are some constraints. Firstly, the model solely considers the building system, rendering it incompatible in certain contexts. Secondly, due to model size limitations, only domestic factors are considered, whereas office and industrial buildings should be incorporated for a more comprehensive analysis [15]. Thirdly, the CO2 emissions calculations only account for operational emissions, neglecting embodied emissions [16,17]. Furthermore, future weather data is not included in the prediction segment due to difficulties in obtaining accurate values. Additionally, the input of the annual change rate and its application to the model may not accurately reflect real-life scenarios, as many factors, such as policy changes and innovations affecting the global energy structure, cannot be easily quantified, or predicted with simple formulas.

Despite the challenges in predicting the future using straightforward values and formulas, emerging tools can be employed to enhance this model. For instance, machine learning can be utilized to predict future weather, behaviour, and other variables by analysing historical data. Another example involves integrating the energy system model with digital twins—a concept that involves creating a real-time, continually updating model of the world—yielding more detailed and precise results for both simulation and prediction.

While the proposed model demonstrates significant potential in urban energy management, several limitations warrant further discussion. Firstly, the exclusion of industrial and office

buildings introduces a critical gap in the model's applicability. Industrial facilities, which often account for 20–30% of urban energy demand, exhibit distinct consumption patterns due to heavy machinery operations, 24/7 operational schedules, and process-specific thermal requirements. For instance, steel manufacturing or data centers demand high-intensity cooling systems, which differ substantially from residential HVAC profiles. By omitting these sectors, the model underestimates peak energy loads and overlooks sector-specific decarbonization opportunities, such as waste heat recovery or industrial symbiosis networks. Future iterations should integrate standardized industrial energy archetypes, leveraging datasets from platforms like the U.S. Department of Energy's Manufacturing Energy Consumption Survey (MECS) or the European Industrial Emissions Portal, to enhance representativeness.

Secondly, the exclusion of embodied emissions—carbon emitted during material extraction, construction, and demolition—limits the model's alignment with lifecycle assessment (LCA) frameworks. For example, modern high-efficiency buildings often utilize carbon-intensive materials like concrete and steel, whose production contributes 8–10% of global CO2 emissions. A holistic urban energy model should incorporate embodied carbon coefficients (e.g., from databases such as Ecoinvent) to evaluate trade-offs between operational efficiency and material sustainability. This integration would enable policymakers to prioritize retrofitting existing structures over new constructions in carbon-sensitive scenarios.

To address prediction uncertainties, machine learning (ML) techniques could be systematically embedded into the model. For instance, recurrent neural networks (RNNs) trained on historical weather and occupancy data could refine future demand forecasts by capturing nonlinear trends and seasonal anomalies. Similarly, reinforcement learning algorithms could optimize real-time energy dispatch strategies for hybrid renewable-grid systems, dynamically balancing supply-demand mismatches. Pilot studies, such as Singapore's "Virtual Singapore" digital twin project, have demonstrated ML's efficacy in reducing prediction errors by 15–20% compared to traditional regression models, underscoring its transformative potential.

The adopted exponential forecasting method, while adequate for this study's preliminary scope, acknowledges limitations in fully internalizing dynamic interdependencies among policy shifts, technological advancements, and climate variations. To address these complexities, future iterations of the model will implement advanced computational frameworks such as deep reinforcement learning (DRL) via Stable Baselines3. This approach aims to dynamically simulate policy impacts (e.g., subsidy reforms or carbon pricing), technological evolution (e.g., breakthroughs in battery storage), and climate adaptation strategies (e.g., extreme weather resilience). By integrating multi-agent systems and real-time feedback loops, DRL could enhance the model's ability to explore optimal energy transition pathways under uncertainty, complementing existing machine learning and digital twin enhancements discussed earlier.

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